

JENKINS ASSIGNMENT

Create a Jenkins Job to Install HTTPS, Start HTTPD, and Deploy index.html

Environment: Jenkins + Amazon Linux EC2 + Apache HTTPD

1. Objective

Create a Jenkins Freestyle Job that connects to the Linux server and performs the following tasks automatically:

- Install Apache HTTPD if it is not already installed.
- Install/configure HTTPS (SSL) for Apache.
- Start and enable the HTTPD service.
- Create and deploy an index.html page.
- Verify that the web server is accessible over HTTP/HTTPS.

2. Prerequisites

- Amazon Linux EC2 instance is running.
- Jenkins is installed and accessible.
- Jenkins job can execute shell commands on the target server.
- Security Group allows SSH (22), HTTP (80), HTTPS (443), and Jenkins (8080) as required.
- You have sudo/root permission on the Linux server.

Note: The commands below are suitable for Amazon Linux 2023. If your assignment server uses another Linux distribution, package names may differ.

3. Architecture / Flow

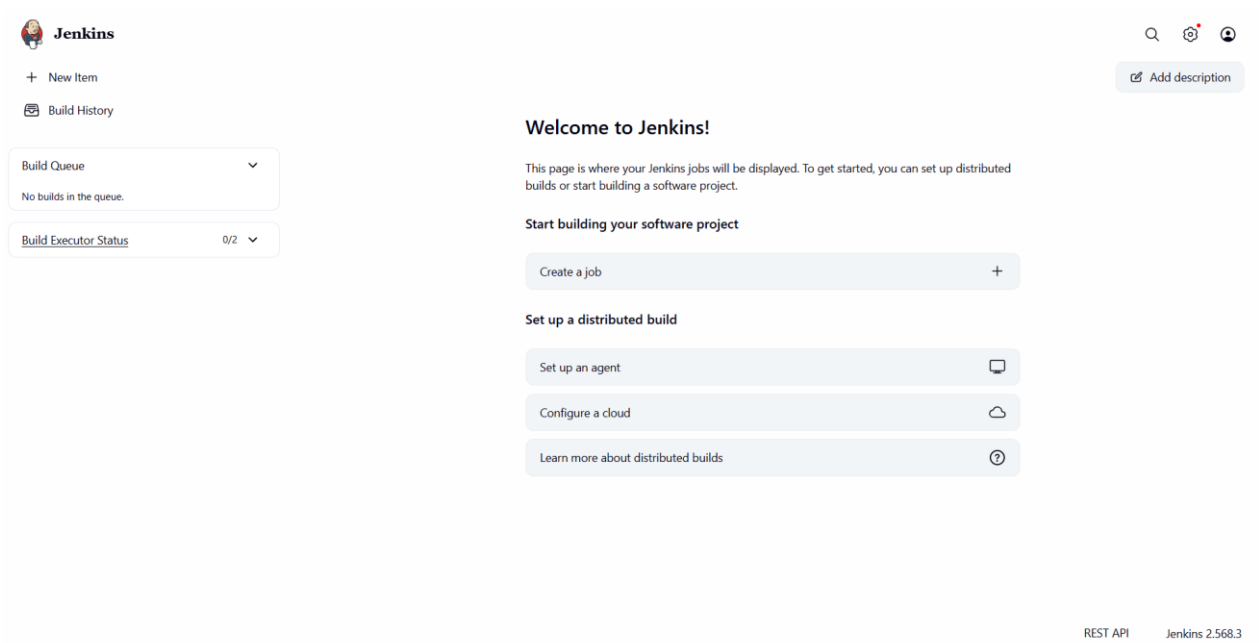
Jenkins → Execute Shell → Linux EC2 → Install HTTPD + SSL → Deploy index.html → Start HTTPD → Verify HTTPS

4. Login to Jenkins

Open Jenkins in your browser, for example:

`http://15.252.97.5:8080/`

Log in using your Jenkins credentials.



5. Create a New Jenkins Job

1. From the Jenkins Dashboard, click New Item.
2. Enter the job name: HTTPD-HTTPS-Deployment
3. Select Freestyle project.
4. Click OK.

← → ↻ 🏠 ⚠ Not secure 15.252.97.5:8080/jenkins/view/all/newJob

🗄 New folder

 **Jenkins** / All ▾ / New Item

New Item

Enter an item name

HTTPD-HTTPS-Deployment_

Select an item type

-  **Pipeline**
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
-  **Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
-  **Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
-  **Folder**
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
-  **Multibranch Pipeline**
Creates a set of Pipeline projects according to detected branches in one SCM repository.
-  **Organization Folder**
Creates a set of multibranch project subfolders by scanning for repositories.

OK

6. Configure the Jenkins Build

In the job configuration page, scroll to Build Steps.

Select: Build Steps → Execute shell

Paste the following script:

```
#!/bin/bash

set -e

echo "==== Installing HTTPD and SSL ====="

sudo dnf install -y httpd mod_ssl openssl

echo "==== Creating index.html ====="

sudo tee /var/www/html/index.html > /dev/null <<'EOF'

<!DOCTYPE html>

<html>

<head>

    <title>Jenkins HTTPD HTTPS Assignment</title>
```

</head>

<body>

<h1>Welcome to My Jenkins Deployment</h1>

<p>HTTPD is running successfully.</p>

<p>HTTPS/SSL is configured.</p>

<p>Deployed through Jenkins.</p>

</body>

</html>

EOF

```
echo "==== Creating SSL certificate ===="
```

```
sudo mkdir -p /etc/pki/tls/private
```

```
sudo mkdir -p /etc/pki/tls/certs
```

```
sudo openssl req -x509 -nodes -days 365 \
```

```
-newkey rsa:2048 \
```

```
-keyout /etc/pki/tls/private/localhost.key \
```

```
-out /etc/pki/tls/certs/localhost.crt \
```

```
-subj "/C=IN/ST=Maharashtra/L=Pune/O=Jenkins/OU=IT/CN=localhost"
```

```
echo "==== Configuring HTTPS ===="
```

```
sudo sed -i 's|^SSLCertificateFile.*|SSLCertificateFile /etc/pki/tls/certs/localhost.crt|' \
/etc/httpd/conf.d/ssl.conf
```

```
sudo sed -i 's|^SSLCertificateKeyFile.*|SSLCertificateKeyFile /etc/pki/tls/private/localhost.key|' \
/etc/httpd/conf.d/ssl.conf
```

```
echo "==== Testing Apache configuration ===="
```

```
sudo apachectl configtest
```

```
echo "==== Starting HTTPD ===="
```

```
sudo systemctl enable --now httpd
```

```
echo "==== HTTPD Status ===="
```

```
sudo systemctl is-active httpd
```

```
echo "==== Testing HTTPS ===="
```

```
curl -k -I https://localhost/
```

```
echo "==== Deployment completed successfully ===="
```

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell ?

Command

See [the list of available environment variables](#)

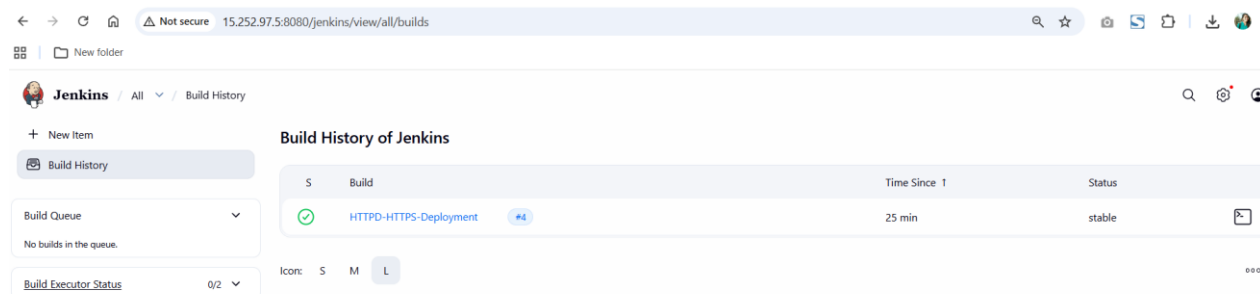
```
#!/bin/bash
set -e

echo "===== Installing HTTPD and SSL ====="
```

Advanced ▾

7. Save and Build the Jenkins Job

1. Click Save.
2. Click Build Now.
3. A build number will appear under Build History.
4. Click the build number.
5. Click Console Output.



8. Verify Console Output

The console should show successful installation, Apache configuration validation, service startup, and HTTPS verification. The final line should be similar to:

===== Deployment completed successfully =====

[illegible]

←

→

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⚠️ Not secure 15.252.97.5:8080/jenkins/job/HTTPD-HTTPS-Deployment/

🗂️

New folder

 **Jenkins** / HTTPD-HTTPS-Deployment

Status

</> Changes

📁 Workspace

▶ Build Now

🗑 Delete Project

⚙ Configure

✎ Rename

Builds > ⋮

🔍 Filter

/

Today

✔️

#4 5:16 pm

▼

✔️ HTTPD-HTTPS-Deployment

Permalinks

- [Last build \(#4\), 24 min ago](#)
- [Last stable build \(#4\), 24 min ago](#)
- [Last successful build \(#4\), 24 min ago](#)
- [Last completed build \(#4\), 24 min ago](#)

9. Verify HTTPD on the EC2 Server

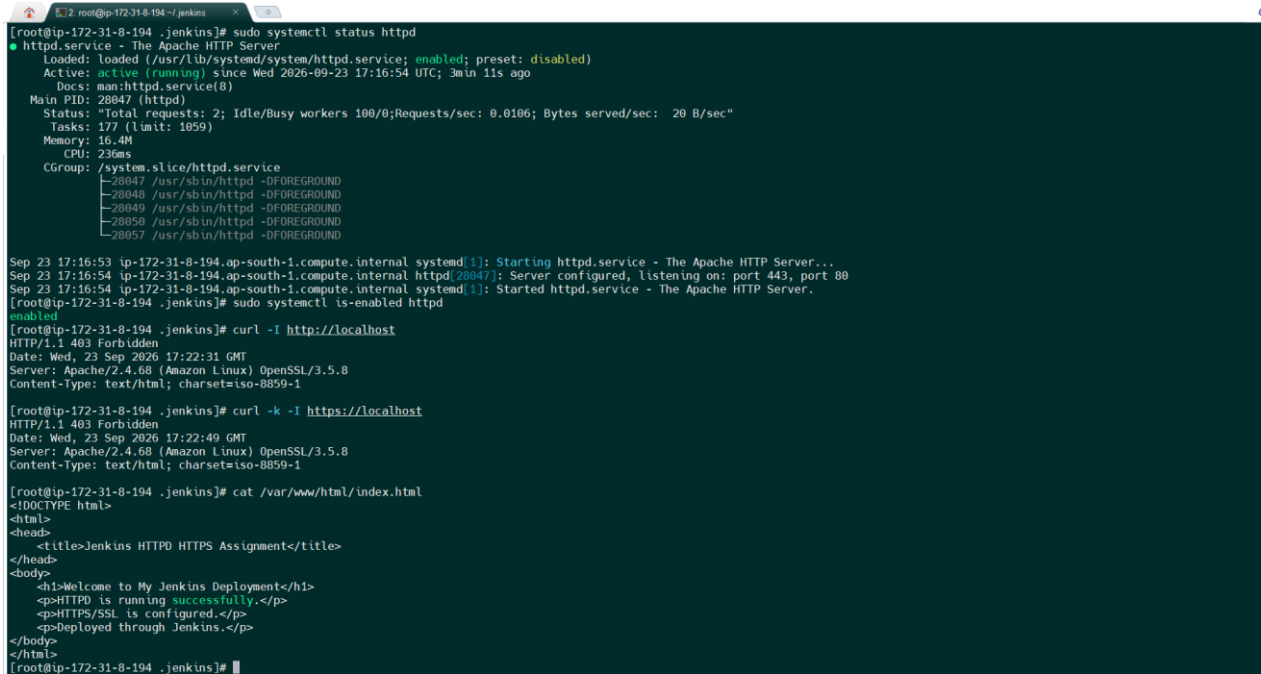
Run these commands on the Linux server:

```
sudo systemctl status httpd
```

```
sudo systemctl is-enabled httpd
```

```
curl -I http://localhost
```

```
curl -k -I https://localhost
```



```
[root@ip-172-31-8-194 .jenkins]# sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Wed 2026-09-23 17:16:54 UTC; 3min 11s ago
     Docs: man:httpd.service(8)
  Main PID: 28047 (httpd)
    Status: "Total requests: 2; Idle/Busy workers 100/0;Requests/sec: 0.0106; Bytes served/sec: 20 B/sec"
   Tasks: 177 (limit: 1059)
  Memory: 16.4M
    CPU: 236ms
  CGroup: /system.slice/httpd.service
          └─28047 /usr/sbin/httpd -DFOREGROUND
            └─28048 /usr/sbin/httpd -DFOREGROUND
              └─28049 /usr/sbin/httpd -DFOREGROUND
                └─28050 /usr/sbin/httpd -DFOREGROUND
                  └─28057 /usr/sbin/httpd -DFOREGROUND

Sep 23 17:16:53 ip-172-31-8-194.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Sep 23 17:16:54 ip-172-31-8-194.ap-south-1.compute.internal httpd[28047]: Server configured, listening on: port 443, port 80
Sep 23 17:16:54 ip-172-31-8-194.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
[root@ip-172-31-8-194 .jenkins]# sudo systemctl is-enabled httpd
enabled
[root@ip-172-31-8-194 .jenkins]# curl -I http://localhost
HTTP/1.1 403 Forbidden
Date: Wed, 23 Sep 2026 17:22:31 GMT
Server: Apache/2.4.68 (Amazon Linux) OpenSSL/3.5.8
Content-Type: text/html; charset=iso-8859-1

[root@ip-172-31-8-194 .jenkins]# curl -k -I https://localhost
HTTP/1.1 403 Forbidden
Date: Wed, 23 Sep 2026 17:22:49 GMT
Server: Apache/2.4.68 (Amazon Linux) OpenSSL/3.5.8
Content-Type: text/html; charset=iso-8859-1

[root@ip-172-31-8-194 .jenkins]# cat /var/www/html/index.html
<!DOCTYPE html>
<html>
<head>
  <title>Jenkins HTTPD HTTPS Assignment</title>
</head>
<body>
  <h1>Welcome to My Jenkins Deployments</h1>
  <p>HTTPD is running successfully.</p>
  <p>HTTPS/SSL is configured.</p>
  <p>Deployed through Jenkins.</p>
</body>
</html>
[root@ip-172-31-8-194 .jenkins]#
```

10. Verify index.html

Run:

```
cat /var/www/html/index.html
```

The deployed HTML content should be displayed.

```
[root@ip-172-31-8-194 .jenkins]# cat /var/www/html/index.html
<!DOCTYPE html>
<html>
<head>
  <title>Jenkins HTTPD HTTPS Assignment</title>
</head>
<body>
  <h1>Welcome to My Jenkins Deployment</h1>
  <p>HTTPD is running successfully.</p>
  <p>HTTPS/SSL is configured.</p>
  <p>Deployed through Jenkins.</p>
</body>
</html>
```

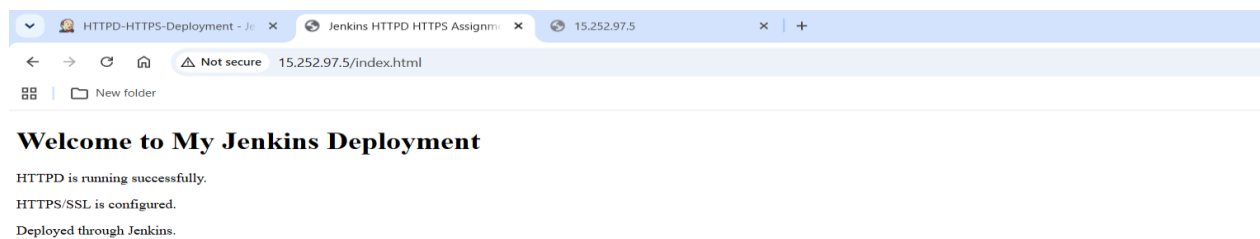
11. Verify Website in Browser

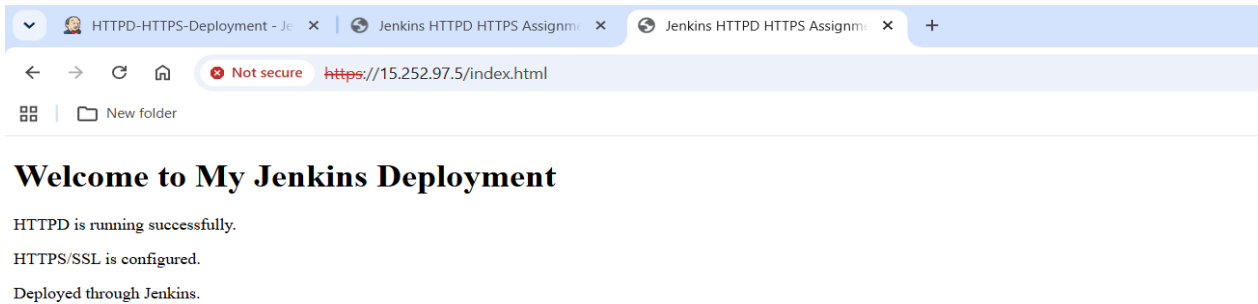
Use the EC2 public IP address:

HTTP: `http://<EC2-PUBLIC-IP>`

HTTPS: `https://<EC2-PUBLIC-IP>`

Because this assignment creates a self-signed certificate, the browser may display a certificate warning. For a lab assignment, proceed to the site after reviewing the warning. A CA-signed certificate should be used for a real production website.

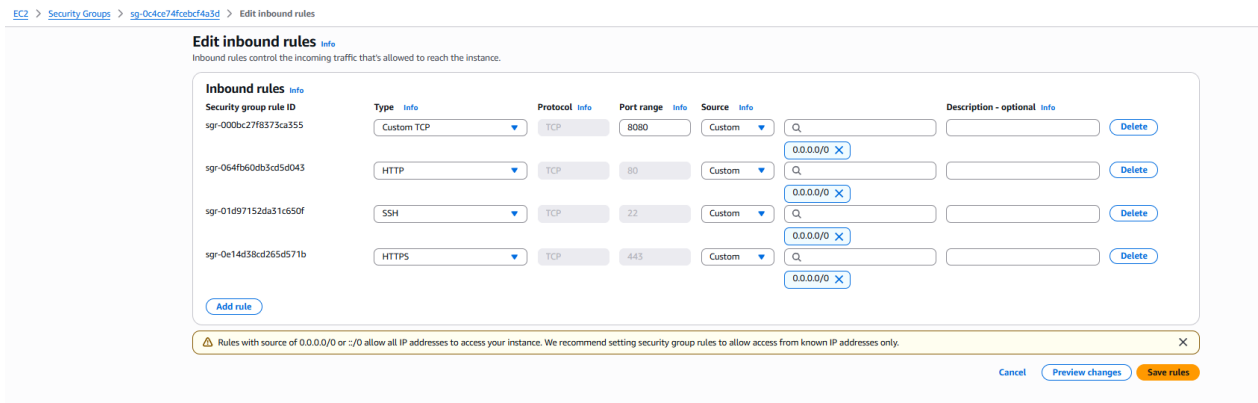




12. AWS Security Group Verification

In the EC2 Security Group, confirm the required inbound rules. Typical lab rules are:

Protocol	Port	Purpose
TCP	22	SSH
TCP	80	HTTP
TCP	443	HTTPS
TCP	8080	Jenkins, if Jenkins is exposed directly



13. Expected Result

The Jenkins job successfully installs Apache HTTPD and SSL packages, creates the HTTPS certificate, deploys index.html, starts/enables HTTPD, and verifies that HTTPS responds successfully.